

19.5 Turing Structures and Propagating Waves

From the delicate beauty of the butterfly to the “fearful symmetry” of the tiger, Nature is full of wondrous patterns, both animate and the inanimate. How do these patterns arise? Dissipative process in systems far from thermodynamic equilibrium may provide at least a partial answer.

The emergence of biological morphology during embryonic development — with hands and feet and eyes all in the right place — is a fascinating subject (a popular account of this subject is Lewis Wolpert’s *Triumph of the Embryo*, 1991, Oxford University Press)). What mechanism produces the morphology of

living organisms? In 1952 the British mathematician Alan Turing suggested a mechanism based on the processes of chemical reactions and diffusion [30]. He showed, by devising a simple model, how chemical reactions and diffusion can work in consonance to produce stable stationary patterns of concentrations. Turing proposed it to explain biological morphogenesis. Today we know that biological morphogenesis is a very complex process, too complex to be explained entirely by the processes of diffusion and chemical reactions. However, Turing's observation has gained much attention since the 1970s due to the great interest in theoretical and experimental study of far-from-equilibrium chemical systems. In this section we will briefly describe a **Turing structure**, or a stationary **spatial dissipative structure**, using the Brusselator of section 19.4.

For simplicity, we shall consider a system with one spatial dimensional, coordinate r , in which diffusion occurs (Fig. 19.10). We assume the system extends from $-L$ to $+L$. We must also specify spatial boundary conditions; the usual boundary conditions are that either the concentrations of the reactants or their flows are maintained at a constant value at the boundaries (or even a combination of both). For our example, we shall assume that the flows of the reactants are zero at the boundaries. Since diffusion flow is proportional to the derivative $\partial C/\partial r$, (in which C is the concentration), the no-flow boundary conditions imply that the derivatives of the concentrations are zero at the boundaries.

When diffusion is included as a transport process, the kinetic equations (19.4.5) and (19.4.6) become

$$\frac{\partial[X]}{\partial t} = D_X \frac{\partial^2[X]}{\partial r^2} + k_1[A] - k_2[B][X] + k_3[X]^2[Y] - k_4[X] \quad (19.5.1)$$

$$\frac{\partial[Y]}{\partial t} = D_Y \frac{\partial^2[Y]}{\partial r^2} + k_2[B][X] - k_3[X]^2[Y] \quad (19.5.2)$$

The boundary conditions are

$$\left. \frac{\partial[X]}{\partial r} \right|_{r=-L} = \left. \frac{\partial[X]}{\partial r} \right|_{r=+L} = 0$$

in which D_X and D_Y are the diffusion coefficients and r is the spatial coordinate. As before, we assume that $[A]$ and $[B]$ are maintained at a fixed uniform value along the entire system (an assumption that simplifies the mathematics but which is difficult to achieve in practice). Diffusion usually homogenizes the concentration in a system, but when coupled with autocatalytic chemical reactions under far-from-equilibrium conditions, it actually generates inhomogeneities or patterns. For pattern formation, the diffusion coefficients must be different. If the diffusion coefficients are nearly equal, then diffusion does not cause an instability; diffusion only tends to homogenize the instability that

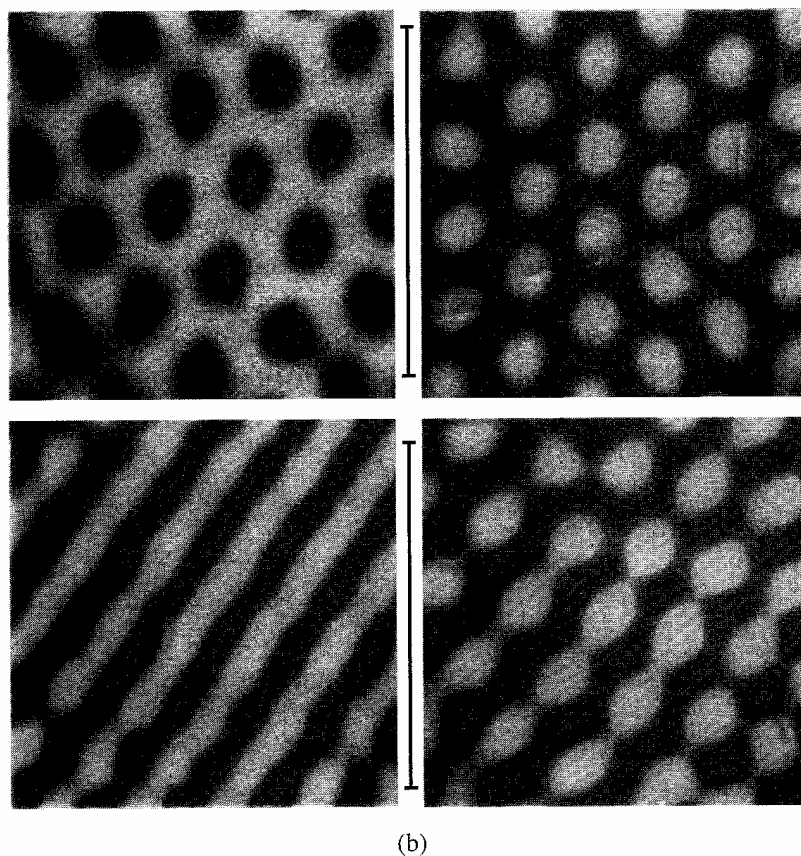
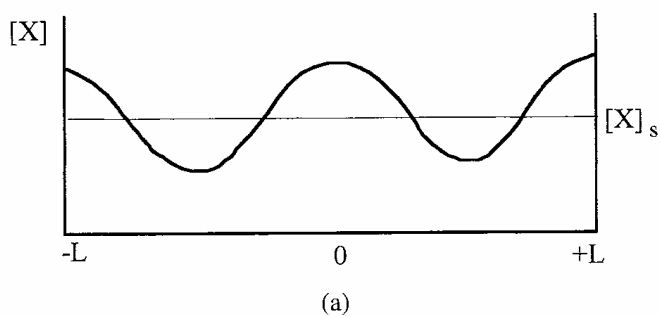


Figure 19.10 (a) Turing structure in a one-dimensional Brusselator model. (b) Turing structures observed in chlorite-iodide-malonic acid reaction in an acidic aqueous solution (Courtesy Harry L. Swinney). The size of each square is nearly 1 mm.

already exists. This can be seen as follows. We begin by considering the stability of the stationary state (19.4.7), the concentrations being homogeneous in the entire system:

$$[X]_s = \frac{k_1}{k_4} [A] \quad [Y]_s = \frac{k_4 k_2}{k_3 k_1} \frac{[B]}{[A]} \quad (19.5.3)$$

The stability of this solution depends on the behavior of a small perturbation. If δX and δY are the small perturbation from $[X]_s$ and $[Y]_s$, it is easy to see that equations linearized about the steady state (19.5.3) are

$$\frac{\partial}{\partial t} \begin{pmatrix} \delta X \\ \delta Y \end{pmatrix} = \begin{bmatrix} D_X \frac{\partial^2}{\partial r^2} & 0 \\ 0 & D_Y \frac{\partial^2}{\partial r^2} \end{bmatrix} \begin{pmatrix} \delta X \\ \delta Y \end{pmatrix} + \begin{bmatrix} k_2[B] - k_4 & k_3[X]_s^2 \\ -k_2[B] & -k_3[X]_s^2 \end{bmatrix} \begin{pmatrix} \delta X \\ \delta Y \end{pmatrix} \quad (19.5.4)$$

If we assume $D_X = D_Y = D$, equation (19.5.4) can be written as

$$\frac{\partial}{\partial t} \begin{pmatrix} \delta X \\ \delta Y \end{pmatrix} = D \frac{\partial^2}{\partial r^2} I \begin{pmatrix} \delta X \\ \delta Y \end{pmatrix} + M \begin{pmatrix} \delta X \\ \delta Y \end{pmatrix} \quad (19.5.5)$$

in which M is the matrix in the second term of (19.5.4) and I is the identity matrix. For a *linear equation* of this type, the spatial part of the solutions can always be written as combinations of $\sin Kx$ and $\cos Kx$, in which the wavenumbers K are chosen so that the boundary conditions are satisfied. This means that if we understand the behavior of a perturbation of the type

$$\begin{pmatrix} \delta X(t) \\ \delta Y(t) \end{pmatrix} \sin Kr \quad \text{and} \quad \begin{pmatrix} \delta X(t) \\ \delta Y(t) \end{pmatrix} \cos Kr \quad (19.5.6)$$

in which the spatial part is separated, then the behavior of all linear combinations of these basic solutions can be deduced. If we substitute (19.5.6) into (19.5.5), we obtain

$$\frac{\partial}{\partial t} \begin{pmatrix} \delta X(t) \\ \delta Y(t) \end{pmatrix} = (-DK^2 I + M) \begin{pmatrix} \delta X(t) \\ \delta Y(t) \end{pmatrix} \quad (19.5.7)$$

From this expression it is clear that if λ_+ and λ_- are the eigenvalues of M , the addition of diffusion will only change the eigenvalues to $(\lambda_+ - DK^2)$ and $(\lambda_- - DK^2)$. Since it is the positivity of the real part of the eigenvalue that indicates instability, we see that (in this case, where $D_X = D_Y$), diffusion does not generate a new instability; it only makes steady states more stable to perturbations with $K \neq 0$. So the solution to (19.5.7) with $K = 0$ is the least stable state because its eigenvalues will have the largest real parts.

For the emergence of spatial patterns, the diffusion coefficients must be *unequal*. In a small region, if one species diffuses out more rapidly than the other, the growth of one species may be facilitated by the depletion of the

other. If this happens, the homogeneous state will no longer be stable and inhomogeneities will begin to grow. When the diffusion coefficients are unequal, it is easy to see that in place of the matrix $(-K^2 DI + M)$ we have the matrix

$$\begin{bmatrix} k_2[B] - k_4 - K^2 D_X & k_3[X]_s^2 \\ -k_2[B] & -k_3[X]_s^2 - K^2 D_Y \end{bmatrix} \quad (19.5.8)$$

For an instability to produce stationary **spatial structures**, the two eigenvalues of this matrix must be real, and at least one must become positive. If the eigenvalues are real, and one becomes positive due to the variations in the parameters $[B]$ and $[A]$, then the unstable perturbation will be of the form

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \sin(Kr) e^{\lambda_+ t} \quad \text{or} \quad \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \cos(Kr) e^{\lambda_+ t} \quad (19.5.9)$$

in which λ_+ is the eigenvalue with positive real part. This indicates a growth of a spatial pattern $\sin Kr$ or $\cos Kr$ without any temporal oscillations; it will evolve to a stationary pattern or a Turing structure.

On the other hand, if the eigenvalues are a complex-conjugate pair, then the solutions to the perturbation equation (19.5.4) will be of the form

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \sin(Kr) e^{(\lambda_{re} \pm i\lambda_{im})t} \quad \text{or} \quad \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \cos(Kr) e^{(\lambda_{re} \pm i\lambda_{im})t} \quad (19.5.10)$$

in which $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$ is the eigenvector with the eigenvalue $\lambda = \lambda_{re} \pm i\lambda_{im}$, with its real and imaginary parts as shown. If the real part λ_{re} is positive, the perturbation (19.5.10) will grow. The unstable perturbation contains oscillations in time, due to the factor $e^{i\lambda_{im}t}$ as well as variations in space due to the factor $\sin Kr$ or $\cos Kr$. Such a perturbation corresponds to a **propagating wave**.

For matrix (19.5.8) the condition for one of its two real eigenvalues to cross zero can be obtained as follows. First we note that the determinant, Det , of a matrix is the product of the eigenvalues. If the eigenvalues are λ_+ and λ_- , we have

$$\begin{aligned} (\lambda_+ \lambda_-) = \text{Det} &= (k_2[B] - k_4 - K^2 D_X)(-k_3[X]_s^2 - K^2 D_Y) \\ &\quad + (k_2[B])(k_3[X]_s^2) \end{aligned} \quad (19.5.11)$$

Before the onset of the instability, both eigenvalues are negative, hence $\text{Det} > 0$. Let us assume that when the parameter $[B]$ is varied, λ_+ crosses zero and becomes positive. Then, at the point where $\lambda_+ = 0$, we have $\text{Det} = 0$ and when $\lambda_+ > 0$ we have $\text{Det} < 0$. Thus the condition for the instability may be stated

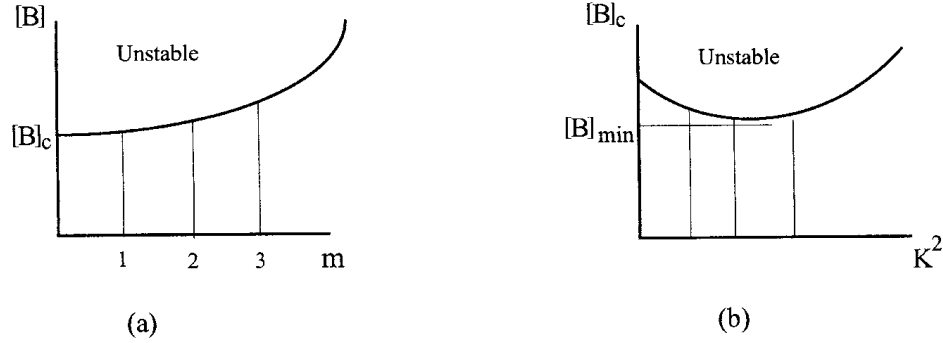


Figure 19.11 (a) Stability diagram showing the value of $[B]$ for which modes of wavenumber m become unstable when D_X and D_Y are nearly equal. The instability occurs when $[B] = [B]_c$ at which $K = 0$, leading to homogeneous oscillations. (b) Stability diagram showing the value of $[B]$ for which modes of wavenumber m become unstable when the difference between D_X and D_Y is large. As $[B]$ increases, when it just above $[B]_{\min}$, the pattern with the value of K^2 that is consistent with the boundary conditions will become unstable and grow

as

$$\begin{aligned} \text{Det} = (k_2 [B] - k_4 - K^2 D_X)(-k_3 [X]_s^2 - K^2 D_Y) \\ + (k_2 [B])(k_3 [X]_s^2) < 0 \end{aligned} \quad (19.5.12)$$

Using $[X]_s = (k_1/k_4) [A]$ this inequality can be rewritten as

$$[B] > \frac{1}{k_2} [k_4 + K^2 D_X] \left[1 + \frac{k_3 k_1^2 [A]^2}{k_4^2} \frac{1}{K^2 D_Y} \right] \quad (19.5.13)$$

This then is the condition under which a Turing structure will arise in the Brusselator model. As $[B]$ increases, the lowest value $[B]_c$ for which (19.5.13) is satisfied will trigger an instability. The value of $[B]_c$ can be found by plotting

$$[B]_c = \frac{1}{k_2} [k_4 + K^2 D_X] \left[1 + \frac{k_3 k_1^2 [A]^2}{k_4^2} \frac{1}{K^2 D_Y} \right] \quad (19.5.14)$$

as a function of K^2 . As shown in Fig. 19.11 (b), this plot has a minimum. When $[B]$ reaches this minimum value, the corresponding $K_{\min}$ will be the wavenumber of the stationary pattern. The minimum occurs at the following values (exc. 19.9):

$$K_{\min}^2 = A \sqrt{\frac{k_3 k_1^2}{k_4 D_X D_Y}} \quad \text{and} \quad [B]_c = [B]_{\min} = \left[\sqrt{k_4} + A \sqrt{\frac{D_X k_3 k_1^2}{D_Y k_4^2}} \right]^2 \quad (19.5.15)$$

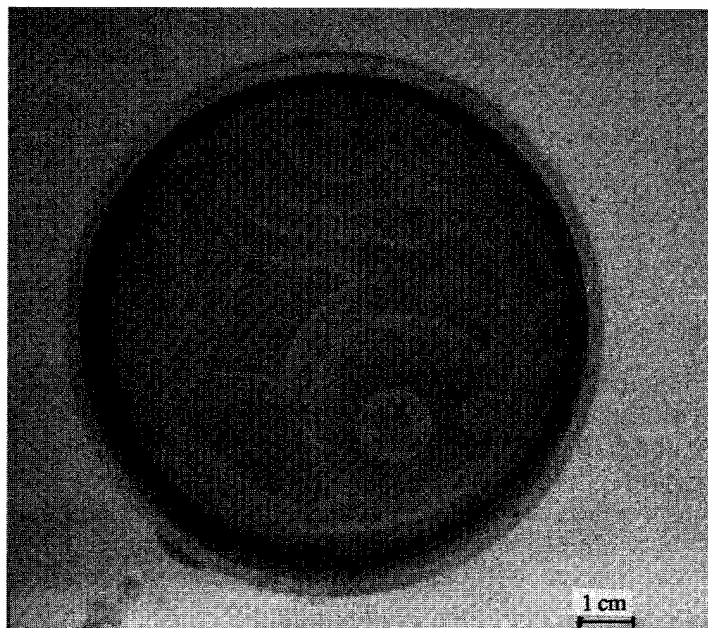


Figure 19.12 Traveling waves in the Belousov–Zhabotinsky reaction

Experimentally, traveling waves have been observed in the Belousov–Zhabotinsky reaction (Fig. 19.12) but only recently have the Turing patterns been realized in the laboratory [31].

The examples shown in this chapter are only a small part of the rich variety of behavior encountered in far-from-equilibrium chemical systems. Here our objective is only to show a few examples; an extensive description would form a book in itself! At the end of the chapter there is a list of monographs and conference proceedings that give a detailed descriptions of oscillations, propagating waves, Turing structures, pattern formation on catalytic surfaces, multistability and chaos (both temporal and spatiotemporal). Dissipative structures have also been found in other fields such as hydrodynamics and optics.